



PAMANTASAN NG LUNGSOD NG SAN PABLO
PATRIOTISM • LEADERSHIP • SERVICE • PROFESSIONALISM

NUMERICAL METHODS

CPE 312

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Lecturer



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Bisection Method



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Learning Objectives

- *Define Bisection Method**
- *Determine and perform the algorithm of Bisection Method**
- *Solve functions using Bisection Method**



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Bisection Method

- alternatively called as binary chopping, interval halving, or Bolzano's method



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Bisection Method

- One type of incremental-search method in which the interval is always divided in half.



Bisection Method

- If a function changes sign over an interval, the function value at the midpoint is evaluated



Bisection Method

- The location of the root is then determined as lying at the midpoint of the subinterval within which the sign change occurs. The process is repeated to obtain refined estimates.



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Bisection Method Algorithm:

For any continuous function $f(x)$,

- Find two points, say a and b such that $a < b$ and $f(a) * f(b) < 0$



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- Find the midpoint of a and b , say " t "
- t is the root of the given function if $f(t) = 0$; else follow the next step



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- Divide the interval $[a, b]$
 - If $f(t) * f(a) < 0$, there exist a root between t and a
 - If $f(t) * f(b) < 0$, there exist a root between t and b



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- Repeat above three steps until $f(t) = 0$.
(stopping criterion 0.00....)



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Illustrative Example # 1:

Determine the root of $f(x) = x^2 - 3$.



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$$f(x) = x^2 - 3$$

$$f(0) = 0^2 - 3 = -3$$



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$$f(x) = x^2 - 3$$

$$f(0) = 0^2 - 3 = -3$$

$$f(1) = 1^2 - 3 = -2$$



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$$f(x) = x^2 - 3$$

$$f(0) = 0^2 - 3 = -3$$

$$f(1) = 1^2 - 3 = -2$$

$$f(2) = 2^2 - 3 = 1$$



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$$f(x) = x^2 - 3$$

$$f(0) = 0^2 - 3 = -3$$

$$f(1) = 1^2 - 3 = -2$$

$$f(2) = 2^2 - 3 = 1$$

$$x \in [1, 2]$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75		



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75	1	



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75	1	0.0625



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75				



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625			



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625	-0.75		



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PATRIOTISM • LEADERSHIP • SERVICE • PROFESSIONALISM

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PATRIOTISM • LEADERSHIP • SERVICE • PROFESSIONALISM

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2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625	-0.75	0.0625	-0.35938



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PATRIOTISM • LEADERSHIP • SERVICE • PROFESSIONALISM

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2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
4	1.625	1.75				



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2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
4	1.625	1.75	1.6875			



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3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
4	1.625	1.75	1.6875	-0.35938	0.0625	-0.15234



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Iterations	a	b	t	f(a)	f(b)	f(t)
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2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
4	1.625	1.75	1.6875	-0.35938	0.0625	-0.15234
5	1.6875	1.75				



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Iterations	a	b	t	f(a)	f(b)	f(t)
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3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
4	1.625	1.75	1.6875	-0.35938	0.0625	-0.15234
5	1.6875	1.75	1.71875			



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2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
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4	1.625	1.75	1.6875	-0.35938	0.0625	-0.15234
5	1.6875	1.75	1.71875	-0.15234	0.0625	-0.04590



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3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
4	1.625	1.75	1.6875	-0.35938	0.0625	-0.15234
5	1.6875	1.75	1.71875	-0.15234	0.0625	-0.04590
6	1.71875	1.75				



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6	1.71875	1.75	1.73438			



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4	1.625	1.75	1.6875	-0.35938	0.0625	-0.15234
5	1.6875	1.75	1.71875	-0.15234	0.0625	-0.04590
6	1.71875	1.75	1.73438	-0.04590	0.0625	0.00807



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75	1	0.0625
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Iterations	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75
2	1.5	2	1.75	-0.75	1	0.0625
3	1.5	1.75	1.625	-0.75	0.0625	-0.35938
4	1.625	1.75	1.6875	-0.35938	0.0625	-0.15234
5	1.6875	1.75	1.71875	-0.15234	0.0625	-0.04590
6	1.71875	1.75	1.73438	-0.04590	0.0625	0.00807

The approximate root of $f(x) = x^2 - 3$ is 1.73438.



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Illustrative Example # 2:

Determine the root of $f(x) = e^{-x} - x$.



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$$f(x) = e^{-x} - x$$

$$f(0) = e^{-0} - 0 = 1$$



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$$f(x) = e^{-x} - x$$

$$f(0) = e^{-0} - 0 = 1$$

$$f(1) = e^{-1} - 1 = -0.63212$$



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$$f(x) = e^{-x} - x$$

$$f(0) = e^{-0} - 0 = 1$$

$$f(1) = e^{-1} - 1 = -0.63212$$

$$x \in [0, 1]$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]



Iterat ions	a	b	t	f(a)	f(b)	f(t)
1	0	1	0.5	1	-0.63212	0.10653
2	0.5	1	0.75	0.10653		



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	0	1	0.5	1	-0.63212	0.10653
2	0.5	1	0.75	0.10653	-0.63212	



Iterat ions	a	b	t	f(a)	f(b)	f(t)
1	0	1	0.5	1	-0.63212	0.10653
2	0.5	1	0.75	0.10653	-0.63212	-0.27763



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	0	1	0.5	1	-0.63212	0.10653
2	0.5	1	0.75	0.10653	-0.63212	-0.27763
3	0.5	0.75				



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	0	1	0.5	1	-0.63212	0.10653
2	0.5	1	0.75	0.10653	-0.63212	-0.27763
3	0.5	0.75	0.625			



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Iterations	a	b	t	f(a)	f(b)	f(t)
1	0	1	0.5	1	-0.63212	0.10653
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Iterations	a	b	t	f(a)	f(b)	f(t)
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3	0.5	0.75	0.625	0.10653	-0.27763	-0.08974



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4	0.5	0.625	0.5625	0.10653	-0.08974	0.00728



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Iterations	a	b	t	f(a)	f(b)	f(t)
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4	0.5	0.625	0.5625	0.10653	-0.08974	0.00728

The approximate root of $f(x) = e^{-x} - x$ is 0.5625.



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ACTIVITY # 4



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Use the bisection method to find the approximate root/s of the following functions:

1. $f(x) = x^3 - 4x - 9$

2. $f(x) = \frac{1}{2}x - \sqrt[3]{x+1}$



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Reminders:

Quiz # 1 (Bring long folder)
Introduction to Numerical Methods
Graphical Method
Incremental-Search Method
Bisection Method



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